

Improvement of thermal stability and electrical performance in HfSiO gate dielectrics by nitrogen incorporation

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ABSTRACT

We have investigated the electrical properties and the thermal stability in terms of crystallization by grazing incidence X-ray diffraction (GI-XRD) and X-ray photoelectron spectroscopy (XPS) in Hf-silicate (HfSiO) and nitrided Hf-silicate (HfSiON) gate dielectric. It is shown that for films with nitrogen incorporation, HfSiON films have superior thermal stability compared to the corresponding HfSiO films. The excellent electrical properties with maximum dielectric constant (17.1 and 18.7) and the smallest oxide-charge density (5.6×10^{11} and $2.2 \times 10^{10} \text{ cm}^{-2}$) and leakage current density (1.3×10^{-6} and $5.9 \times 10^{-7} \text{ A/cm}^2$ at $V_g = -1 \text{ V}$) were obtained after 900 and 950°C annealing for HfSiO and HfSiON films, respectively, which indicates that the electrical performance of HfSiO gate dielectrics had a larger improvement through nitrogen incorporation.

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1. Introduction

As the aggressive scaling of Si-based complementary metal oxide semiconductors (CMOS) has reached fundamental physical limitations, it requires metal gate electrodes and high-K gate dielectrics to replace the conventional polycrystalline silicon gate electrode and SiO₂/SiON gate dielectrics for the purpose of alleviating their excessive gate leakage current and polycrystalline silicon gate depletion [1–4]. Among the candidates of high-K materials, HfSiO(N) has drawn particular attention because of its high thermal stability, excellent interfacial properties, and appropriate band alignment with Si [5–8]. Also, some reports have addressed additional advantages of nitrided Hf-silicate such as minimized interfacial layer formation [9], reduced boron diffusion from the poly-Si gate into the channel [10], which can increase dielectric constant and improve reliability [11,12].

However, there are several problems in their particular use, such as the formation of metallic component, crystallization, and oxygen vacancy during high temperature annealing for dopant activation progress [13–16]. In this paper, we have investigated the influence of annealing on crystallization, thermal stability, and electrical characteristics of HfSiO and HfSiON gate dielectrics.

2. Experimental

P-type Si(1 0 0) substrates with a resistivity of 1–10 Ωcm were bombarded using assisted source ion beam from dual ion beam sputtering deposition (DIBSD) for 10 min, in order to remove native oxide and contaminations. HfSiO films were prepared by DIBSD using main source ion beam to bombard compound targets (a Si (99.99%, $\phi = 100 \text{ mm}$) disk covered with a small HfO₂ (99.99%, $\phi = 60 \text{ mm}$) disk at room temperature). While nitrogen atoms were incorporated into the HfSiO films using assisted source ion beam to bombard substrate, HfSiO films were depositing. Argon (99.999%) and nitrogen (99.999%) were used as main source and assisted source gas, respectively. The Si/Hf composition ratio was controlled via changing the relative location of HfO₂ target attached on the Si target and NaCl wafer was used to determine the Si/Hf composition ratio by energy dispersive X-ray spectroscopy (EDXS). Details about the principle of the main components of the DIBSD system were reported in Ref. [17]. MOS capacitors were formed with 50 nm-thick Au top and bottom electrodes. The electrodes were prepared by RF sputtering at room temperature through a shadow mask with a diameter of 0.1 mm; the actual electrode area of each capacitor was measured with an optical microscope. All MOS capacitors were annealed at 400 °C for 10 min to form a good ohmic contact.

Film thickness was estimated from cross-sectional scanning electron microscope (Hitachi S-4700) for all samples. A Rigaku D/max-3C X-ray diffraction system was used for the grazing

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incidence X-ray diffraction (GI-XRD) to detect crystallization. Hitachi S-4700 energy dispersive X-ray spectroscopy (EDXS) and phi-500X-ray photoelectron spectroscopy (XPS) systems were used to determine the chemical composition and chemical bonding state. 1 MHz capacitance–voltage ($C-V$) and current–voltage ($I-V$) measurements were measured with HP4294A and Keithley 6517, respectively.

3. Result and discussion

3.1. Thermal stability characteristics

GI-XRD spectra of as deposited HfSiO(N) films and for films annealed at different temperature in oxygen atmosphere for 30 min are displayed in Fig. 1(a) and (b). The physical thickness of both as deposited films was ~ 30 nm. [Hf] content ($[Hf]/([Hf]+[Si])$) was measured by EDXS (the HfSiO(N) films on NaCl substrate were used

for EDXS analysis). As shown in the spectra, both films deposited were amorphous. On comparison of Fig. 1(a) and (b), XRD shows crystallization temperatures of HfSiO film with 20.2% [Hf] content and HfSiON film with 20.2% [Hf] content and 32.44% [N] content are 950 °C and 1050 °C, respectively, which are much higher than 500 °C from that of m-HfO₂ [18]. It was shown that HfSiO(N) film had better thermal stability than HfO₂ film. Pant et al. [19] pointed out that the activation energy for HfSiO continuous nucleation and growth appears to be high due to the constraint of SiO₂. Another possible reason is that the HfSiO film consists of more Si–O bonding than HfO₂ film, which may appear to suppress crystallization. Within the same [Hf] content, crystallization temperature of HfSiON films increased from 950 to 1050 °C, which indicated that the thermal stability of the dielectric film is improved with nitrogen-incorporated HfSiO films.

In order to further demonstrate the improvement of thermal stability with nitrogen incorporation, the Hf4f core-level photoelectron spectra from X-ray photoelectron spectroscopy (XPS) are shown in Fig. 2 for (a) HfSiO film with 20.2% [Hf] content and 0% [N] content

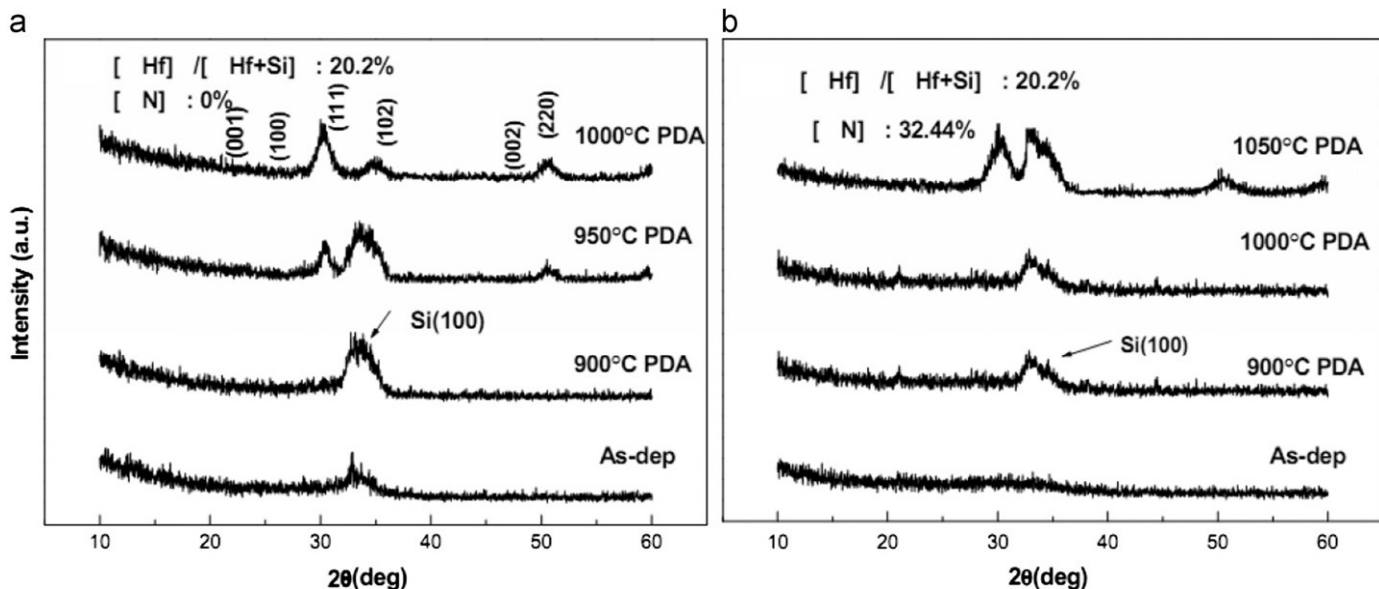


Fig. 1. GI-XRD spectra with different annealing temperatures for (a) HfSiO and (b) HfSiON films.

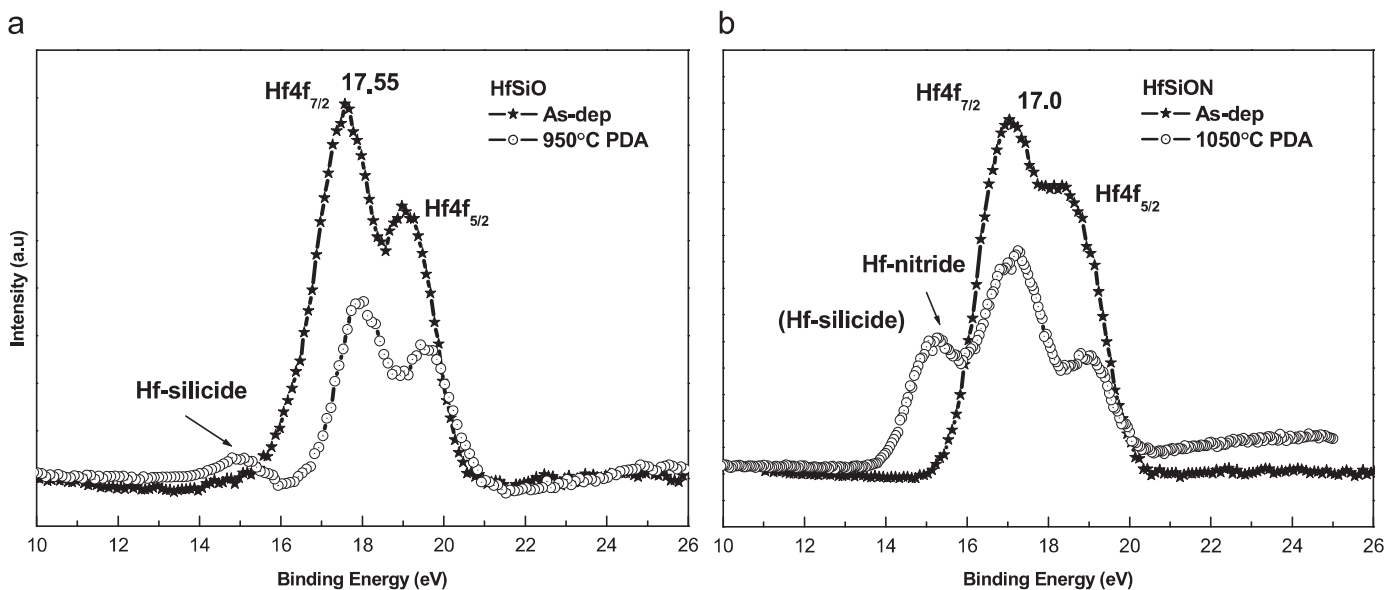


Fig. 2. Annealing-temperature dependence of Hf 4f core-level photoelectron spectra for (a) HfSiO with $[Hf]/([Hf]+[Si])=20.2\%$, $[N]=0\%$ and (b) HfSiON films with $[Hf]/([Hf]+[Si])=20.2\%$, $[N]=32.44\%$.

and (b) HfSiON film with 20.2% [Hf] and 32.44% [N] content. In order to meet the XPS test requirements for film thickness, Ar-ion stripping method was used, and the XPS results were obtained when the film thickness was less than 2 nm. The measured binding energies (BE) were calibrated by the peak energy of C1s (285.0 eV). As can be seen, as-deposited films exhibit a doublet due to spin-orbit splitting. The Hf4f_{7/2} peaks at 17.55 and 17.0 for HfSiO and HfSiON film, respectively, originate from Hf bound to oxygen [20]. For both films with annealing at 950 and 1050 °C, Hf4f peak could be convoluted into three peaks, corresponding to Hf4f_{7/2}, Hf4f_{5/2}, and Hf-silicide (~14.8 eV) or Hf-nitride (~15.2 eV). While Hf-silicide may slightly coexist in the metallic Hf-nitride components due to the spectra of Hf4f metallic component are quite broader than those of the Hf-silicide [21]. Silicidation action occurs at different annealing temperatures of 950 and 1050 °C for HfSiO and HfSiON films, respectively, suggesting that the incorporation of nitrogen indeed enhances the film thermal stability.

Fig. 3 shows the annealing-temperature dependence of O1s and Si2p core-level photoelectron spectra for (a, c) HfSiO and (b, d) HfSiON films. As shown in Fig. 3(a) and (b), for both films as deposited, O–Si bonding (~532.5 eV) is more dominant than O–Hf bonding (~530.1–531 eV). As the annealing temperature increases, the HfSiON film shows a drastic increase in the intensity of O–Hf bonding as compared to that of HfSiO film, indicating that the formation of O–Si bonding has been inhibited at the bottom interface due to the N atoms being substituted by the O atoms. In Fig. 3(c) and (d), by annealing at 850 and 900 °C, respectively, the intensity of Si oxide component increases for both films compared to the as-deposited sample, indicating that the oxidation of Si substrate occurs at the dielectric/Si substrate interface [22]. These results from O1s and Si2p core-level spectra suggest that the incorporation of nitrogen suppresses the continuous growth of SiO layer between the HfSiO(N) film and the Si substrate [23].

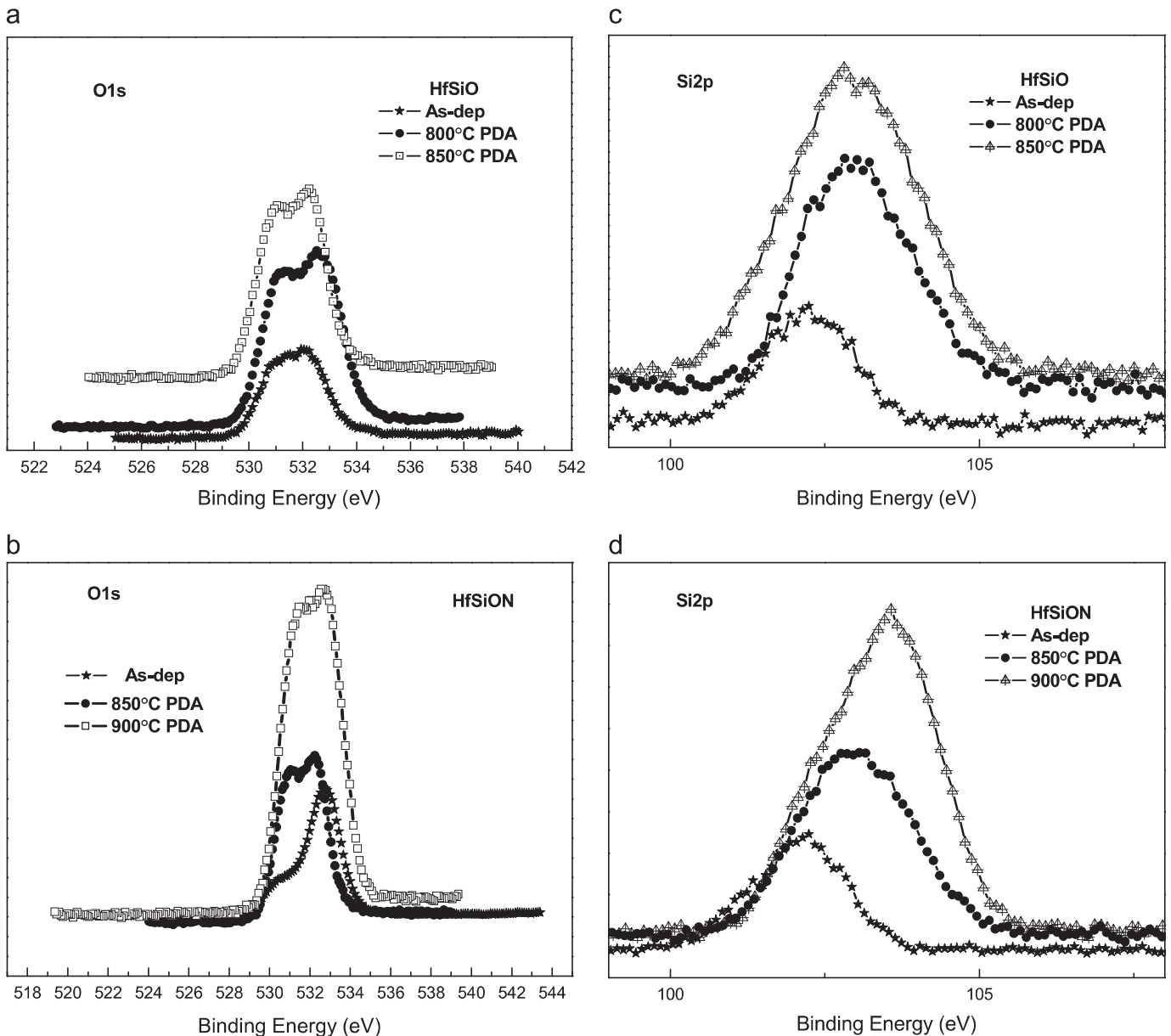


Fig. 3. Annealing-temperature dependence of O1s and Si2p core-level photoelectron spectra for (a, c) HfSiO with [Hf]/([Hf] + [Si]) = 20.2%, [N] = 0%, and (b, d) HfSiON films with [Hf]/([Hf] + [Si]) = 20.2%, [N] = 32.44%.

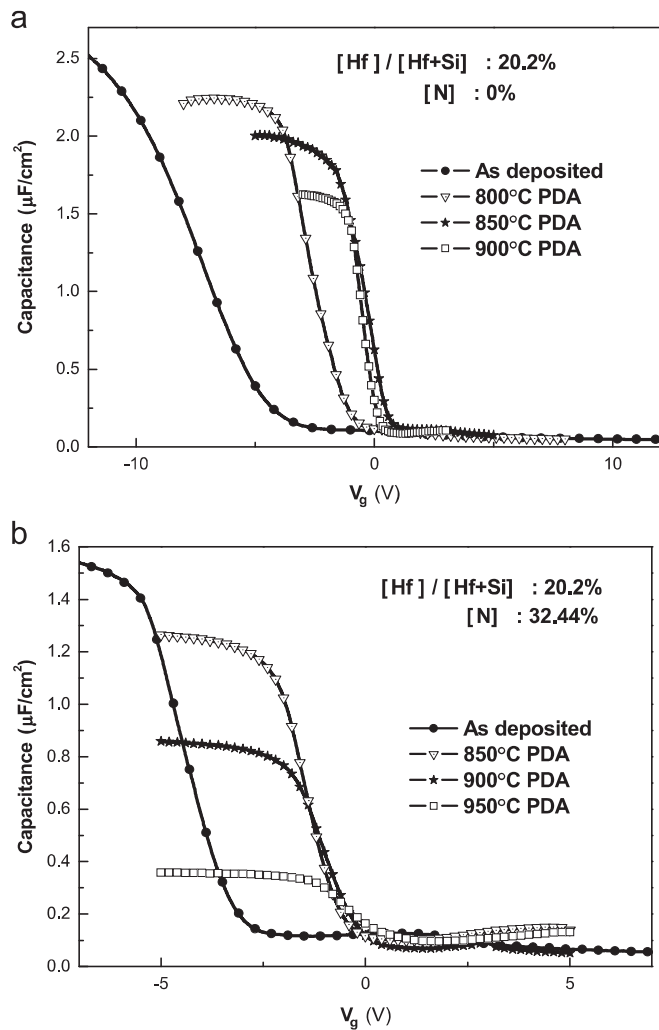


Fig. 4. High-frequency (1 MHz) C–V curves of (a) HfSiO capacitors with $[Hf]/([Hf]+[Si])=20.2\%$, $[N]=0\%$ and (b) HfSiON capacitors with $[Hf]/([Hf]+[Si])=20.2\%$, $[N]=32.44\%$ capacitors at different annealing temperatures.

3.2. C–V characteristics

High-frequency (1 MHz) C–V curves of HfSiO($[Hf]/[Hf+Si]=20.2\%$, $[N]=0\%$) and HfSiON ($[Hf]/[Hf+Si]=20.2\%$, $[N]=32.44\%$) capacitors are presented in Fig. 4(a) and (b) for structures annealed at different temperatures. An initial hysteresis of about 0.6 and 0.36 V (not shown) for the as-deposited HfSiO and HfSiON films, respectively, is reduced to about 0.18 and 0.05 V after annealing at 900 and 950 °C, respectively. Post annealing reduction in the hysteresis width, determined as the flat band voltage shift in the C–V characteristics measured from depletion to accumulation and vice versa, is attributed to the decreasing role of charge carrier trap releasing processes [24].

Subsequent changes in the flat band voltage (V_{fb}) and the equivalent oxide-charge density (Q_{ox}), which were extracted from the C–V curves for HfSiO and HfSiON films, are presented in Fig. 5. V_{fb} is related to Q_{ox} , which including fixed charges, border-trap charges, mobile-ion charges, and interface-state charges is calculated according to $Q_{ox} = -C_{ox}(V_{fb} - \phi_{ms})/q$, where ϕ_{ms} is the work-function difference between Au and Si substrates, and C_{ox} is the oxide capacitance per unit area. For annealing temperatures up to 950 °C, the shifts in V_{fb} are 6.22 and 3.75 V, and correspondingly Q_{ox} changes from 3.2×10^{12} to $5.6 \times 10^{11} \text{ cm}^{-2}$ and from 8.1×10^{11} to $2.2 \times 10^{10} \text{ cm}^{-2}$ for HfSiO and HfSiON films, respectively. For the HfSiO film, large Q_{ox} implies that a large amount of defects are generated near the interface, which is due to the generation of SiO layer at the bottom interfaces between the interfacial silicate layers and the Si substrates [25,26]. For the HfSiON films, incorporated nitrogen plays a role of compensating the oxygen vacancies accumulated, which have been suggested as a possible source of charge trap, at the lower interface in the HfSiO layer [27,28]. Therefore, nitrogen incorporation into HfSiO films may effectively suppress the SiO layer formation at the interface due to a decrease in the number of oxygen vacancies. Additionally, it has been reported that there is a distinct correlation between crystallization and charge trapping in HfSiON films [29,30]. One possible reason for the reduction in charge trapping in HfSiON films may be a result of increase in the crystallinity of films due to the increased thermal stability, which had been investigated in the previous section.

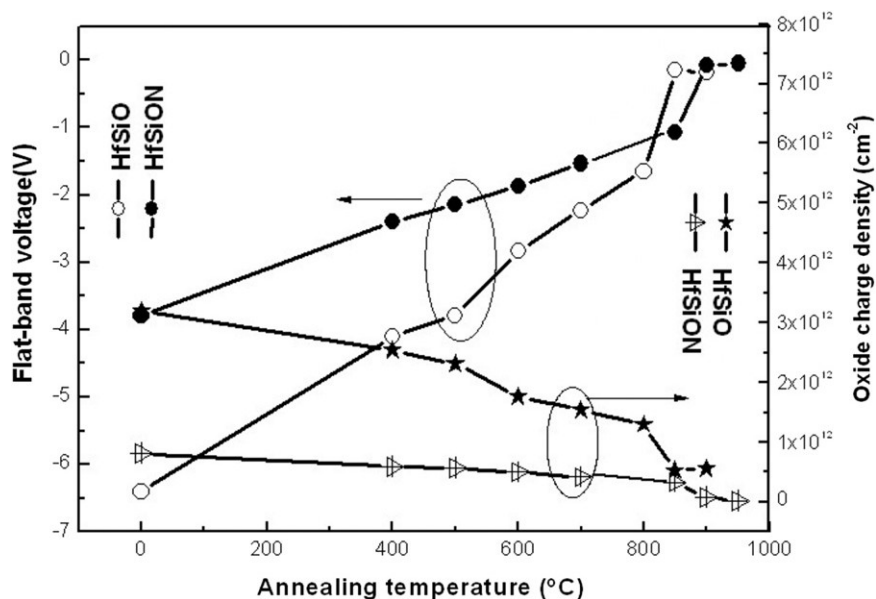


Fig. 5. Flat band voltage and oxide charge density of HfSiO and HfSiON films vs annealing temperature.

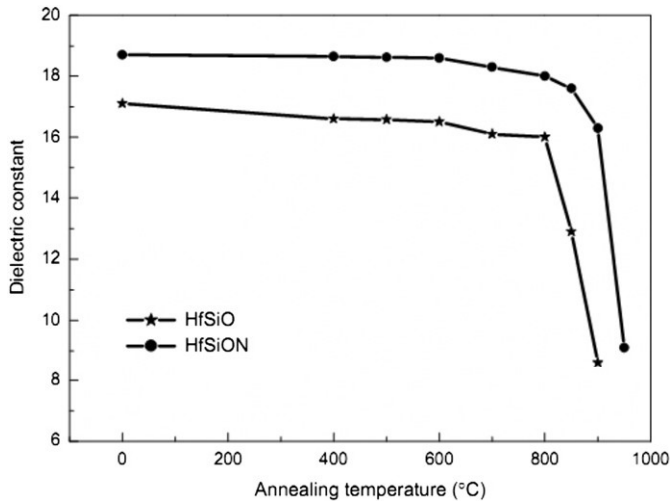


Fig. 6. Change with different annealing temperatures of the dielectric constant for HfSiO and HfSiON films.

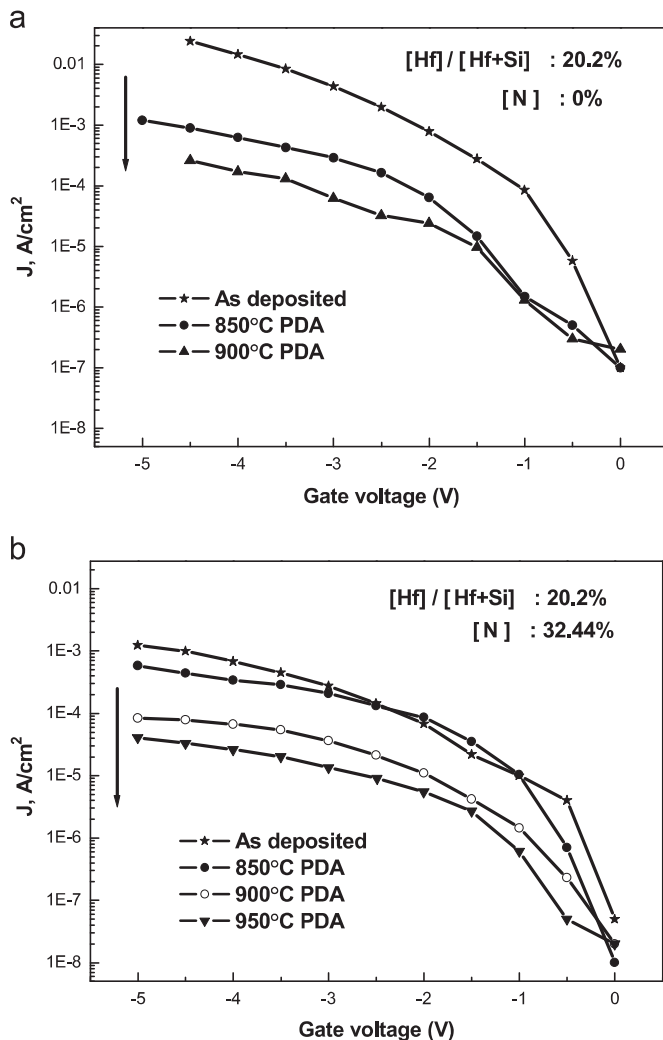


Fig. 7. Gate leakage current density vs gate voltages for (a) HfSiO and (b) HfSiON films at different annealing temperatures.

Fig. 6 shows complementary information on the dielectric constant. A decrease in the dielectric constant with the increase of annealing temperature is observed here for both films.

The maximum dielectric constants of 17.1 and 18.7 are obtained for as deposited HfSiO and HfSiON films, respectively. Several effects take place in the interface layers above 800 °C: (1) oxygen atoms diffuse into the Si substrate, forming SiO_x clusters, which increase the lower interface layers thickness, and is similar to XPS analysis shown in Fig. 3. (2) Si atoms diffuse into the top layers resulting in a metal-silicate type composite. All these processes reduce the overall dielectric constant of both films.

Another interesting observation is the higher dielectric constant (by approximately 10–15%) for the HfSiON films. One reason for the observation was suggested in Ref. [31]: the change in trap density with annealing temperature, which manifests itself in the hysteresis of the C–V curves. Another possible reason is that the ion polarity of chemical bond is enhanced due to the replacement of part oxygen atoms by nitrogen atoms, leading to increase of dielectric constant in HfSiON films.

3.3. I–V characteristics

Gate leakage current is another important device parameter. To evaluate the gate leakage performance, the J – V curve is measured in accumulation region, as shown in Fig. 7(a) and (b). High density of bulk defects and oxide charges density could be responsible for the leakage properties. There is a large reduction of leakage current (two- and one-orders of magnitude) at 900 and 950 °C PDA for HfSiO and HfSiON films, respectively, compared to that of as-deposited of both films. As can be seen from Fig. 7(a), HfSiO film after 900 °C annealing exhibited the smallest leakage current density ($1.3 \times 10^{-6} \text{ A/cm}^2$) at $V_g = -1 \text{ V}$, which was attributed to lower oxide-charge density from Fig. 4. While in Fig. 7(b), HfSiON film after 950 °C annealing showed the minimum values ($5.9 \times 10^{-7} \text{ A/cm}^2$) at $V_g = -1 \text{ V}$, which is one order of magnitude lower than that of HfSiO film after 900 °C annealing. It is because adding nitrogen to the gate dielectrics can reduce leakage current in the films [32].

4. Conclusion

In summary, we have investigated the thermal stability of HfSiO and HfSiON films annealed at different temperatures by GI-XRD and XPS; the thermal stability of the dielectric film was enhanced with the incorporation of nitrogen atoms incorporated. Electrical performance of HfSiO gate dielectrics had an obvious improvement through nitrogen incorporation. The maximum dielectric constant increased from 17.1 to 18.7 for as deposited and the smallest leakage current density and Q_{ox} decreased from 1.3×10^{-6} to $5.9 \times 10^{-7} \text{ A/cm}^2$ at $V_g = -1 \text{ V}$ and from 5.6×10^{11} to $2.2 \times 10^{10} \text{ cm}^{-2}$ after 950 °C annealing, were obtained for HfSiO gate dielectrics through nitrogen incorporation.

Acknowledgments

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